SVM →

Support Vector Machines

Agenda ⇒

Maths Heavy ⇒

- SVMs
- Hard margin
- soft margin
- Hinge loss
- intuition
- kernel Trick

Code ⇒

Sklearn

Train, Test

SVM → supervised algo (X, Y)

→ classification | Regression

→ 1990s 2000s

V. popular

SVM

imagenet competitions

Power → KERNELIZATION

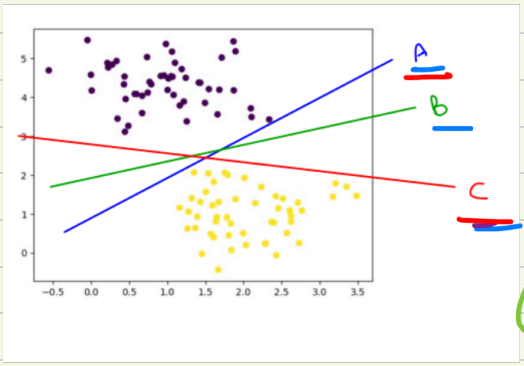
Best

Maths

SVM

DATA size ✗

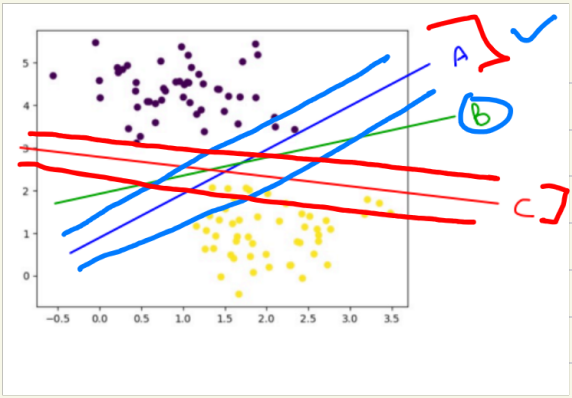
BINARY classification



100% accuracy

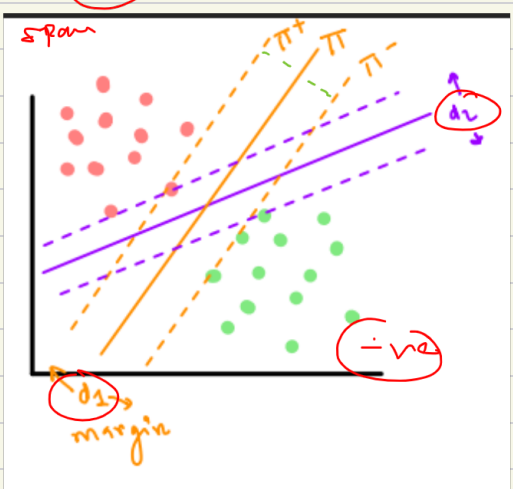
Best → SVM

Correct predn / Total no. of pts ✗



→ Dp very close to the margin

SVM → margin ↑



idea → the hyperplane should maximize the margin

maximized

maximize ?

92%

98%

→ more confidence in pred^n

clear separation

↳ Better generalization

margin = maximizing classifier

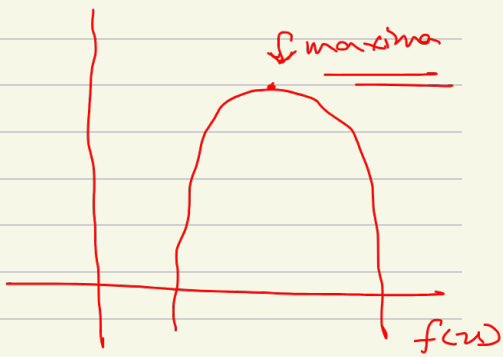
margin maximizing classifier

** Obj func^n → line, plane, Hyperplane which can maximize the margin

↓

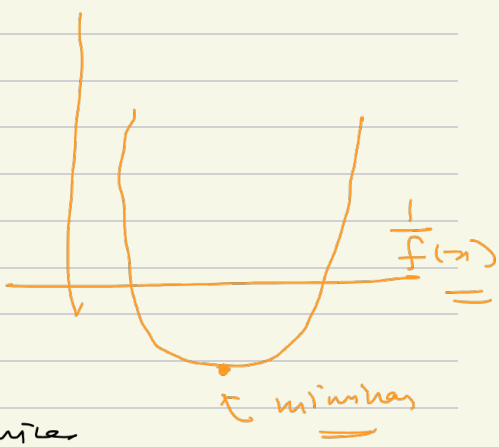
optimization

** argmax [hyperplane which maximizes the margin]



**

argmin [1 / hyperplane which max. margin]



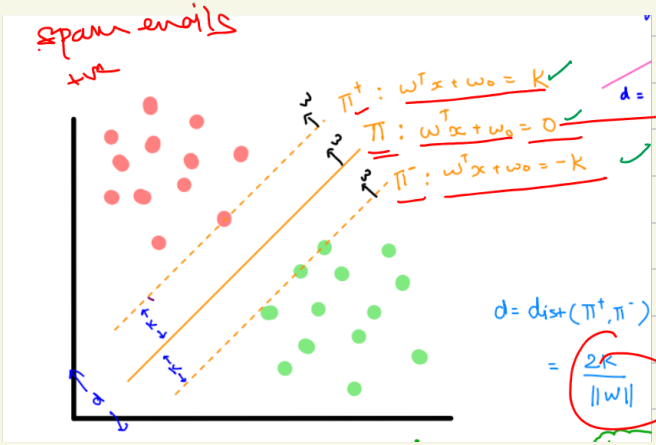
x

5 = 1/5

maximize = minimize

Minimization problem

(4D)



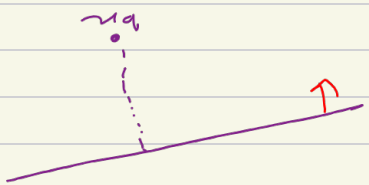
→ main hyperplane

parallel

$\vec{w}$ same

w_0 ≠

$$\text{dist. B/w } (\pi^+, \pi^-) = \frac{2k}{\|w\|}$$



$$d = \frac{w^T x_q + w_0}{\|w\|}$$

$\frac{2k}{\|w\|}$ ↑ ↓

$k = 1$

Calculations simpler

case 1

$k = 1$

$d = \frac{2(1)}{\|w\|}$

$\underset{w, w_0}{\text{argmax}} \frac{2}{\|w\|}$

$\downarrow$

$\underset{w, w_0}{\text{argmin}} \frac{\|w\|}{2}$

case 2

$k = 10$

$d = \frac{20}{\|w\|}$

$\underset{w, w_0}{\text{argmax}} \frac{20}{\|w\|}$

$\downarrow$

$\underset{w, w_0}{\text{argmin}} \frac{\|w\|}{20}$

$\Rightarrow$

20 & 2 same

$f(x) = (x-5)^2 \Rightarrow x = 5$

$f'(x) = 2 \cdot (x-5)^2 + 10 = 0$

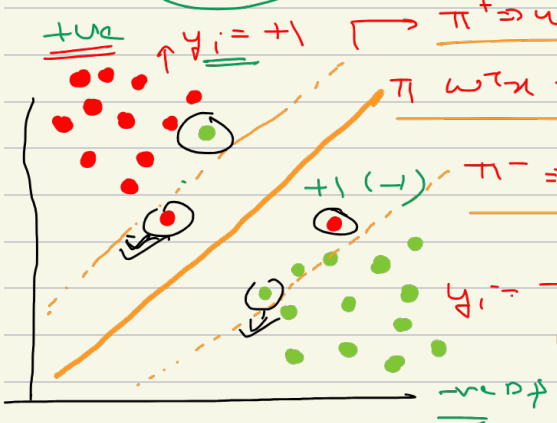
$x = 5$

$k = k \Rightarrow k = 1$

obj func^n

$\underset{w, w_0}{\text{argmax}} \frac{2}{\|w\|}$ or $\underset{w, w_0}{\text{argmin}} \frac{\|w\|}{2}$

w, w_0 → optimizing



-ve -ve → +ve

-ve -ve > 1

obj func^n

$\underset{w, w_0}{\text{argmin}} \frac{\|w\|}{2}$

such that / every DP should be on their correct side

$\Rightarrow$ Hard constraints

Practically impossible

SVM → Hard margin

Strict constraints

Hard margin ✗

Quiz time!

Time Left: 35s

What do you mean by generalization in terms of SVM?

20 users have participated

A	How far the hyperplane is from the training datapoints	40%
✓ B	How accurately the SVM can predict outcomes for unseen data	35%
C	How accurately the SVM classifies training datapoints	25%

End Quiz Now

unseen data

Train

Test

Generalization

Quiz time!

Time Left: 30s

What do you mean by a hard margin?

27 users have participated

✓ A	The SVM allows no error in classification.	81%
B	The SVM allows some error in classification.	15%
C	The SVM allows high error in classification.	4%

End Quiz Now

Hard margin SVM

y_i → actual target

Obj. func^n

$$\underset{w, w_0}{\operatorname{argmin}} \frac{\|w\|}{2}$$

constraint

such that

$$y_i \cdot (w^T x_i + w_0) \geq 1$$

$\forall i : 1 \to N$

⇒ DP should be on their correct region

log Reg^n

y_i : 0 → +1

1 → +1

log loss

SVM

y_i : +1

−1

Obj. fun^n

$$\underset{w, w_0}{\arg\min} \frac{\|w\|}{2},$$

such that $y_i \cdot (w^T x_i + w_0) \geq 1$

$\forall i : 1 - N$

How many constraints

N constraints

+ve sample

$y_i = +1$

$w^T x_i + w_0 \geq 1$

+ve ≥ 1

$=$ Hard margin $=$

-ve sample

$y_i = -1$

$w^T x + w_0 \leq -1$

+ve' $\gg$

10:07

$\leq$ (min) 10:07

SV $\rightarrow +1$ & -1

Hard margin

Soft margin → soft heart

some dp to have small errors

soft margin

added to obj. func

all dp have errors ?

NO

soft margin

Real world → X

Hard margin → tiny constraint

Soft margin →

Balance b/w margin maximi ratio

v/s loss that does not affect margin

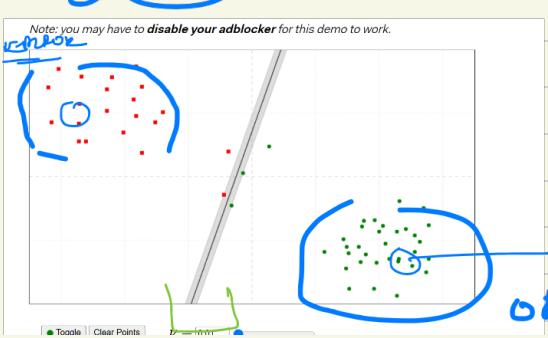
Hinge loss

obj →

$$\underset{w, w_0}{\arg\min} \frac{\|w\|}{2} + (\text{error}_i)$$

$$\max(0, 1 - z_i)$$

k=1

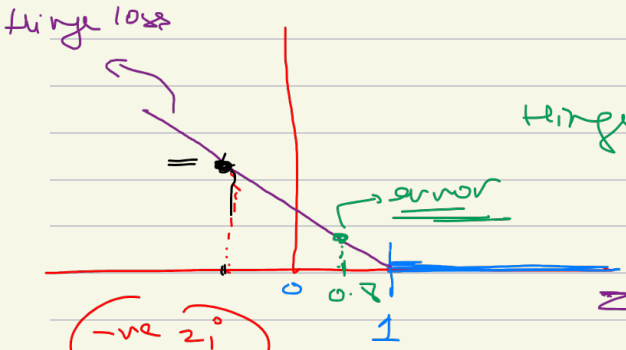


$z_i \rightarrow y_i \cdot (w^T x_i + w_0)$

2 D

adding errors → margin is maximized

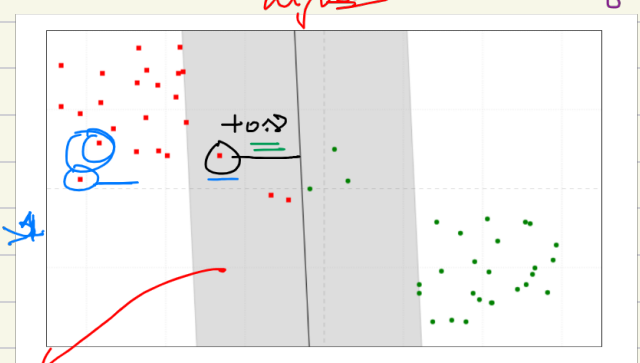
Hinge loss



Hinge loss

$z_i = y_i \cdot (w^T x_i + w_0)$

penalty will be higher



if $z_i > 1$

$\xi_i = 0$

penalty

not $z_i > 1$

$y_i = +1$

$y_i \ (0.8)$

error $\max(0, 0.8)$

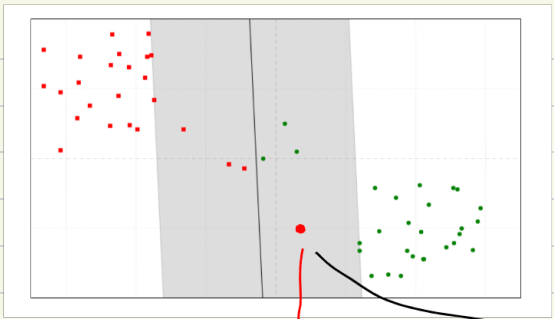
$\Rightarrow 0.8$

$z_i \rightarrow y_i$ $+1$ 0.8

$\xi_i = \max(0, 1 - 0.8)$

$\max(0, 0.2)$

ξ_i 0.2



$z_i = -ve$

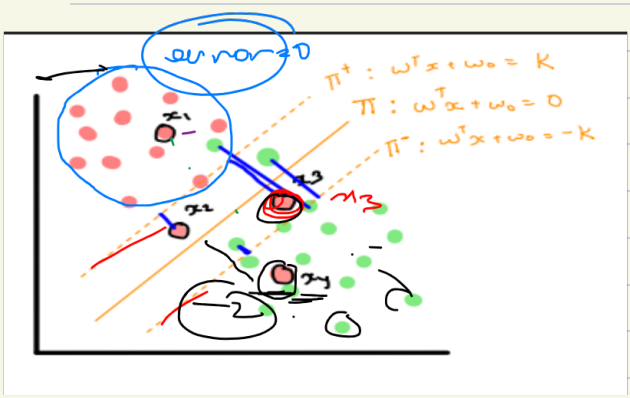
$y_i = +1$, $d_i = -ve$

$z_i = y_i \cdot d_i = -ve$

$e_i = \max(0, 1 - z_i)$

$1 - (-z_i)$

$(1 + z_i)$ Belongs



$x_1 \rightarrow z_1$

$z_1 = y_1 (w^T x_1 + w_0)$

$+1 \quad 2$

$= +2$

$e_1 = \max(0, 1 - 2)$

$\max(0, -1)$

$0 = \text{error}$

$x_2 \rightarrow z_2$

$z_2 = y_2 \cdot (w^T x_i + w_0)$

0.5

$z_2 = 0.8$

$e_i = \max(0, 1 - 0.5) = 0.5$



$x_3 \rightarrow z_3$

$z_3 = (+1)(-0.5)$

$= \quad = -0.5$

$e_i = \max(0, 1-(-0.5))$

$= \max(0, 1.5)$

$= 1.5$ error

$x_4 \rightarrow z_4$

$z_4 = y_i (w^T x_i + w_0)$

$(+1)(-2) = -2$

$e_i = \max(0, 3)$

$e_i = 3$ large error



Hy

combine

generalizing

hyperparam

$$\underset{w, w_0}{\operatorname{argmin}} \left[\frac{\|w\|}{2} + C \cdot \sum_{i=1}^{N} \max(0, 1 - z_i) \right]$$

penalty :

such that

$$z_i = y_i (w^T x_i + w_0)$$

error ≥ 0

Soft margin

C↑ C → Hy per param in SVM

C = 0

C = 109

⇒ underfit

more imp to error / loss part

(0.1) (0.2)

0.1 × 109

Hard margin

a lot of penalty

⇒ overfitting

Regularization

Reg.

C = 1/λ

→ Regularization

λ ↑ ⇒ underfit

C ↑ = overfitting

hard margin

λ ↓ → overfit

C ↓ = underfit

What's the use of 'C' here?

C is a hyperparameter.
It controls whether to focus on :

- Maximizing margin
- Minimizing errors ξ_i 's

Case 1

As C ↓

- Model becomes too lenient towards misclassifications.
- More importance is given to maximizing margins.
- The model may Underfit.

Case 2

As C ↑

- Model becomes less lenient towards misclassifications.
- More importance is given to minimizing errors.
- The model may Overfit.

$$\underset{w, w_0}{\arg\min} \frac{\|w\|}{2} + C \cdot \frac{1}{n} \sum_{i=1}^{n} \xi_i$$

such that $y_i (w^T x_i + w_0) \geq 1 - \xi_i$

primal form of obj fun

hinge

≥ 1

$(1 - 0.2)$

0.2 unit

1 DP

2nd DP → error = 20

3rd → error = 2.9

Research

Primal Dual equivalence

Primal → Change → Dual form

Dual form → w, wo

Extend

Primal Dual Equivalence

$$\underset{\alpha_i}{\operatorname{argmax}} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \cdot \alpha_j \cdot y_i y_j \, x_i^T x_j$$

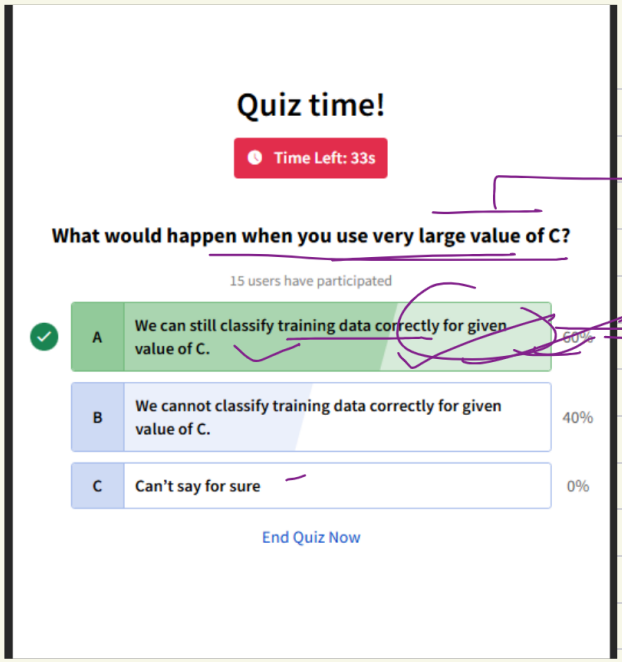
s.t $0 \le \alpha \le C$

$\sum_{i=1}^{N} \alpha_i y_i = 0$

$\forall_{i:1 \to N}$

→ find α_i'

Dual advantage


Quiz time!
Time Left: 33s
What would happen when you use very large value of C?
15 users have participated
A
We can still classify training data correctly for given value of C.
60%
B
We cannot classify training data correctly for given value of C.
40%
C
Can't say for sure
0%
End Quiz Now